8. A number consists of two digits whose sum is 9. If 27 is subtracted from the original number, its digits are interchanged. Find the original number.
9. The denominator of a fraction exceeds its numerator by 8. If the numerator is increased by 17 and the denominator is decreased by 1, we get $\frac{3}{2}$. Find the original fraction.
10. If a train runs at 60 km/hr it reaches its destination late by 15 minutes. But, if it runs at 85 kmph it is late by only 4 minutes . Find the distance covered by the train.

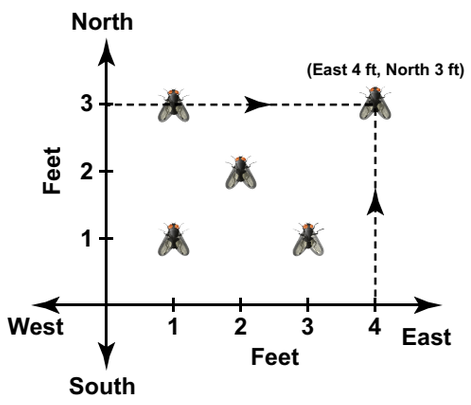
Objective Type Questions

11. Sum of a number and its half is 30 then the number is_____.
    (A) 15 (B) 20 (C) 25 (D) 40
12. The exterior angle of a triangle is 120° and one of its interior opposite angle 58°, then the other opposite interior angle is_____.
    (A) 62° (B) 72° (C) 78° (D) 68°
13. What sum of money will earn ₹500 as simple interest in 1 year at 5% per annum?
    (A) 50000 (B) 30000 (C) 10000 (D) 5000
14. The product of LCM and HCF of two numbers is 24. If one of the number is 6, then the other number is _____.
    (A) 6 (B) 2 (C) 4 (D) 8
15. The largest number of the three consecutive numbers is x+1, then the smallest number is
    (A) x (B) x+1 (C) x+2 (D) x–1

3.9 Graph

3.9.1 Introduction

There was an instance in the 17th century when **Rene Descartes**, a famous mathematician became ill and from his bed, noticed an insect hovering over a corner and sitting at various places on the ceiling. He wanted to identify all the places where the insect sat on the ceiling. Immediately, he drew the top plane of the room in a paper, creating the horizontal and vertical lines. Based on these perpendicular lines, he used the directions and understood that the places of the insect can be spotted by the movement of the insect in the east, west, north and south directions. He called that place as (x,y) in the plane which indicates two values, one (x) in the horizontal direction and the other (y) in the vertical direction (say east and north in this case). This is how the concept of graphs came into existence.



3.9.2 Graph sheets

Graph is just a visual method for showing relationships between numbers. In the previous class, we studied how to represent integers on a number line horizontally. Now take one more number line – vertically. We take the graph sheet keeping both number lines mutually perpendicular to each other at '0'zero as given in the figure. The number lines and the marking integers should be placed along the dark lines of the graph sheet.

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